Unit 3: Stratification
Stratified Random Sampling

A. Purpose of stratification
B. Notation
C. Inference
D. Precision & design effect

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A. Purpose of stratification

- Example: Estimate the mean energy consumption of commercial buildings in a state (no. of kW-h)
- Frame with N = 5,318 commercial buildings

Building ID	Location	Size	…
1	Area 1	580,200	
2	Area 1	45,630	
⋮	⋮	⋮	
5,318	Area 23	360,380	

- Objectives:
  - Control the distribution of sample buildings by size (square footage)
  - Improve the precision over SRS

SRS review

- Objective, randomized selection
- Every element has an equal chance of selection
- Every possible sample has the same probability of being drawn
- Even "bad" samples have an equal chance of being selected
  - "Bad samples": a break down of the randomization principle
- Further, other sample designs can achieve smaller sampling error for the same cost
- Or same sampling error for lower cost

Stratified sampling

- Objective:
  - Can we sample to avoid, or even eliminate, "bad" samples?
  - Can we achieve smaller sampling error for the same cost or same sampling error for lower cost?
- Stratification uses information other than labels (i.e., $i = 1, 2, \ldots, N$)
  - "Auxiliary information" used to create strata
  - Group elements into H mutually exclusive and exhaustive groups (strata)
  - Select samples from each stratum
  - Compute estimates for each stratum, and combine

Stratified sampling: Example

- Control distribution of the sample buildings by size (square footage):

Stratum	Group	Count
1	< 50K sq. ft.	1,182
2	50 - 250K	1,846
3	250 - 4,000K	1,395
4	>= 4,000K	895
Total		5,318

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B. Notation

- Notation for stratified sampling is same as that for SRS, with a subscript to denote strata
- For the population *N* elements in *H* strata with N_h elements in stratum *h*:
  - $\sum_{h=1}^{H} N_h = N_1 + N_2 + \ldots + N_H = N$
  - $Y_{h1}, Y_{h2}, \ldots, Y_{hN_h},\ h = 1, \ldots, H$

Population level

- For each stratum *h*:

$$Y_h = \sum_{i=1}^{N_h} Y_{hi}$$

$$\bar{Y}_h = \frac{Y_h}{N_h} = \frac{1}{N_h} \sum_{i=1}^{N_h} Y_{hi}$$

$$S_h^2 = \frac{1}{N_h - 1} \sum_{i=1}^{N_h} \left(Y_{hi} - \bar{Y}_h \right)^2$$

- For dichotomous variables:

$$\bar{Y}_h = P_h \qquad S_h^2 = \frac{N_h}{N_h - 1} P_h \left(1 - P_h \right)$$

Sample level

- Sample sizes n_h and sample elements $y_{h1}, y_{h2}, \ldots, y_{hn_h}$:

$$y_h = \sum_{i=1}^{n_h} y_{hi}$$

$$\bar{y}_h = \frac{y_h}{n_h} = \frac{1}{n_h} \sum_{i=1}^{n_h} y_{hi}$$

$$\sum_{h=1}^{H} n_h = n_1 + n_2 + \ldots + n_H = n$$

$$s_h^2 = \frac{1}{n_h - 1} \sum_{i=1}^{n_h} \left(y_{hi} - \bar{y}_h \right)^2$$

- For dichotomous variables:

$$\bar{y}_h = p_h \qquad s_h^2 = \frac{n_h}{n_h - 1} p_h \left(1 - p_h \right)$$

Example: Stratified sample data, *n* = 500

Stratum h	Group	N_h	n_h	$\bar{y}_h$	s_h^2
1	< 50K sq. ft.	1,182	100	72.7	11,783.6
2	50 - 250K	1,846	200	336.1	205,657.8
3	250 - 4,000K	1,395	120	1,690.2	3,568,970.7
4	>= 4,000K	895	80	11,659.0	183,039,437.2
Total		5,318	500		

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C. Inference

- At population level,

$$\bar{Y} = \frac{\sum_{h=1}^{H}\sum_{i=1}^{N_h} Y_{hi}}{N} = \frac{\sum_{h=1}^{H} Y_h}{N} = \frac{\sum_{h=1}^{H}\left(\frac{N_h}{N_h}\right) Y_h}{N} = \frac{\sum_{h=1}^{H} N_h \left(\frac{Y_h}{N_h}\right)}{N} =$$

$$\sum_{h=1}^{H} \frac{N_h}{N} \bar{Y}_h = \sum_{h=1}^{H} W_h \bar{Y}_h$$

- At sample level,

$$\bar{y}_w = \sum_{h=1}^{H} W_h \bar{y}_h$$

Element level weighting

- The stratum level weighted mean can be expressed with element level weights:

$$\bar{y}_w = \sum_{h=1}^{H} W_h \bar{y}_h = \sum_{h=1}^{H} \left(\frac{N_h}{N} \right) \bar{y}_h = \frac{1}{N} \sum_{h=1}^{H} N_h \frac{y_h}{n_h} = \frac{1}{N} \sum_{h=1}^{H} \frac{N_h}{n_h} \sum_{i=1}^{n_h} y_{hi} =$$

$$= \frac{1}{N} \sum_{h=1}^{H} \sum_{i=1}^{n_h} \left(\frac{N_h}{n_h} \right) y_{hi} = \frac{1}{N} \sum_{h=1}^{H} \sum_{i=1}^{n_h} w_{hi} y_{hi} = \frac{\sum_{h=1}^{H} \sum_{i=1}^{n_h} w_{hi} y_{hi}}{\sum_{h=1}^{H} \sum_{i=1}^{n_h} w_{hi}}$$

- Here,

$$\sum_{h=1}^{H} \sum_{i=1}^{n_h} w_{hi} = \sum_{h=1}^{H} \sum_{i=1}^{n_h} \left(\frac{N_h}{n_h} \right) = \sum_{h=1}^{H} n_h \left(\frac{N_h}{n_h} \right) = \sum_{h=1}^{H} N_h = N$$

Sampling variance

- Variances are combines across strata

$$\begin{aligned}
\mathrm{Var}\left(\bar{y}_w\right) &= \mathrm{Var}\left(\sum_{h=1}^{H} W_h \bar{y}_h\right) \\
&= \mathrm{Var}\left(W_1\bar{y}_1 + W_2\bar{y}_2 + \cdots + W_H\bar{y}_H\right) \\
&= \mathrm{Var}\left(W_1\bar{y}_1\right) + \mathrm{Var}\left(W_2\bar{y}_2\right) + \cdots + \mathrm{Var}\left(W_H\bar{y}_H\right) \\
&= W_1^2\mathrm{Var}\left(\bar{y}_1\right) + W_2^2\mathrm{Var}\left(\bar{y}_2\right) + \cdots + W_H^2\mathrm{Var}\left(\bar{y}_H\right) \\
&= \sum_{h=1}^{H} W_h^2 \mathrm{Var}\left(\bar{y}_h\right)
\end{aligned}$$

- Estimate this sampling variance with $\mathrm{var}\left(\bar{y}_w\right) = \sum_{h=1}^{H} W_h^2 \mathrm{var}\left(\bar{y}_h\right)$

SRS within strata: Stratified Random Sampling

- Adding strata subscript, $\operatorname{var}\left(\bar{y}_h\right)=\dfrac{1-f_h}{n_h}s_h^2$
- And combining across strata:

$$\operatorname{var}\left(\bar{y}_w\right)=\sum_{h=1}^{H}W_h^2\operatorname{var}\left(\bar{y}_h\right)=\sum_{h=1}^{H}W_h^2\frac{1-f_h}{n_h}s_h^2$$

- For proportions,

$$\operatorname{var}\left(p_w\right)=\sum_{h=1}^{H}W_h^2\left(\frac{1-f_h}{n_h-1}\right)p_h\left(1-p_h\right)$$

Confidence Intervals

- Confidence intervals:

$$\bar{y}_w \pm t_{1-\alpha/2,df}\sqrt{\text{var}(\bar{y}_w)} = \bar{y}_w \pm t_{1-\alpha/2,df}\sqrt{\sum_{h=1}^{H} W_h^2 \frac{1-f_h}{n_h} s_h^2}$$

$$p_w \pm t_{1-\alpha/2,df}\sqrt{\text{var}(p_w)} = p_w \pm t_{1-\alpha/2,df}\sqrt{\sum_{h=1}^{H} W_h^2 \left(\frac{1-f_h}{n_h-1}\right) p_h\left(1-p_h\right)}$$

$$\text{where } df = \sum_{h=1}^{H}\left(n_h-1\right) = n-H$$

Example: Stratified sample data, *n* = 500

Stratum h	Group	N_h	n_h	$\bar{y}_h$	s_h^2
1	< 50K sq. ft.	1,182	100	72.7	11,783.6
2	50 - 250K	1,846	200	336.1	205,657.8
3	250 - 4,000K	1,395	120	1,690.2	3,568,970.7
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Total		5,318	500		

$$\bar{y}_w = \sum_{h=1}^{H} W_h \bar{y}_h = W_1\bar{y}_1 + W_2\bar{y}_2 + W_3\bar{y}_3 + W_4\bar{y}_4 =$$

$$= \frac{1,182}{5,318} \times 72.7 + \frac{1,846}{5,318} \times 336.1 + \frac{1,395}{5,318} \times 1,690.2 + \frac{895}{5,318} \times 11,659.0 = 2,538.4$$

Example: Stratified sample data, *n* = 500

Stratum h	Group	N_h	n_h	$\bar{y}_h$	s_h^2
1	< 50K sq. ft.	1,182	100	72.7	11,783.6
2	50 - 250K	1,846	200	336.1	205,657.8
3	250 - 4,000K	1,395	120	1,690.2	3,568,970.7
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Total		5,318	500		

$$\mathrm{var}\left(\bar{y}_w\right)=\sum_{h=1}^{H}W_h^2\,\mathrm{var}\left(\bar{y}_h\right)=\sum_{h=1}^{H}W_h^2\frac{1-f_h}{n_h}s_h^2=$$

$$=\left[\left(\frac{1{,}182}{5{,}318}\right)^2\times\frac{1-\dfrac{100}{1{,}182}}{100}\times 11{,}783.6\right]+\ldots+\left[\left(\frac{895}{5{,}318}\right)^2\times\frac{1-\dfrac{80}{895}}{80}\times 183{,}039{,}437.2\right]=60{,}998.0$$

Example: Stratified sample data, *n* = 500

Stratum h	Group	N_h	n_h	$\bar{y}_h$	s_h^2
1	< 50K sq. ft.	1,182	100	72.7	11,783.6
2	50 - 250K	1,846	200	336.1	205,657.8
3	250 - 4,000K	1,395	120	1,690.2	3,568,970.7
4	>= 4,000K	895	80	11,659.0	183,039,437.2
Total		5,318	500		

$$\bar{y}_w \pm t_{1-0.05/2,n-H}\sqrt{\mathrm{var}(\bar{y}_w)} = 2{,}538.4 \pm 1.96\sqrt{60{,}998.0} =$$

$$= \left[2{,}053.1\,;\,3{,}023.6\right]$$

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D. Precision & design effect

- What can we gain from this more complicated method of sample selection and estimation?
  - Better control, administrative convenience
  - Reduced variance – in some allocations
- How can we measure such potential gains in precision?
  - Compare sampling variance of the stratified estimate against its corresponding SRS sampling variance – Design effect
- Total variance in the population:

$$(N-1)S^2 = \sum_{h=1}^{H}\sum_{i=1}^{N_h}\left(Y_{hi} - \bar{Y}\right)^2$$

Analysis of variance

- Expanding the total variance

$$\sum_{h=1}^{H}\sum_{i=1}^{N_h}\left(Y_{hi}-\bar{Y}\right)^2=\sum_{h=1}^{H}\sum_{i=1}^{N_h}\left(Y_{hi}-\bar{Y}_h+\bar{Y}_h-\bar{Y}\right)^2$$

$$=\sum_{h=1}^{H}\sum_{i=1}^{N_h}\left[\left(Y_{hi}-\bar{Y}_h\right)+\left(\bar{Y}_h-\bar{Y}\right)\right]^2$$

$$=\sum_{h=1}^{H}\sum_{i=1}^{N_h}\left[\left(Y_{hi}-\bar{Y}_h\right)\right]^2+\sum_{h=1}^{H}\sum_{i=1}^{N_h}\left[\left(\bar{Y}_h-\bar{Y}\right)\right]^2$$

$$+2\sum_{h=1}^{H}\sum_{i=1}^{N_h}\left[\left(Y_{hi}-\bar{Y}_h\right)\left(\bar{Y}_h-\bar{Y}\right)\right]$$

Components of variance

- Further simplification leads to…

$$\sum_{h=1}^{H}\sum_{i=1}^{N_h}\left(\frac{N_h-1}{N_h-1}\right)\left(Y_{hi}-\bar{Y}_h\right)^2+\sum_{h=1}^{H}N_h\left(\bar{Y}_h-\bar{Y}\right)^2$$

$$=\sum_{h=1}^{H}\left(N_h-1\right)S_h^2+\sum_{h=1}^{H}N_h\left(\bar{Y}_h-\bar{Y}\right)^2$$

$$S^2=\sum_{h=1}^{H}\frac{\left(N_h-1\right)}{\left(N-1\right)}S_h^2+\sum_{h=1}^{H}\left(\frac{N_h}{N-1}\right)\left(\bar{Y}_h-\bar{Y}\right)^2$$

$$\approx\sum_{h=1}^{H}W_hS_h^2+\sum_{h=1}^{H}W_h\left(\bar{Y}_h-\bar{Y}\right)^2$$

$$=\textit{Within Variance}+\textit{Between Variance}$$

Designing stratified samples

- The sampling variance for the mean of a stratified sample is based on the within component only
  - Between stratum variance is ignored completely
- Decrease the sampling variance of the mean by making strata heterogeneous between and homogeneous within
- Use auxiliary *X* correlated with *Y*, since *Y* unknown
- Gains are not guaranteed
  - Depends on allocation of sample to strata - values of n_h

Design effect

- Gains (or losses) in precision can be measured through the design effect (*deff*):

$$deff = \frac{\operatorname{var}\left(\bar{y}_w\right)}{\operatorname{var}_{SRS}\left(\bar{y}\right)} = \frac{\sum_{h=1}^{H} W_h^2 \left(\frac{1-f_h}{n_h}\right) s_h^2}{\left(\frac{1-f}{n}\right) s^2}$$

- Gain in precision if *deff* < 1
- Losses in precision *deff* > 1

Example: Stratified sample data, *n* = 500

Stratum h	Group	N_h	n_h	$\bar{y}_h$	s_h^2
1	< 50K sq. ft.	1,182	100	72.7	11,783.6
2	50 - 250K	1,846	200	336.1	205,657.8
3	250 - 4,000K	1,395	120	1,690.2	3,568,970.7
4	>= 4,000K	895	80	11,659.0	183,039,437.2
Total		5,318	500		

$$deff = \frac{\mathrm{var}\left(\bar{y}_w\right)}{\mathrm{var}_{SRS}\left(\bar{y}\right)} = \frac{\sum_{h=1}^{H} W_h^2 \frac{1-f_h}{n_h} s_h^2}{\frac{1-f}{n} s^2} = \frac{60,988.0}{\frac{1-\frac{500}{5,318}}{500} \times 49,038,472.8} = 0.6865$$

$$s^2 \approx \sum_{h=1}^{H} W_h s_h^2 + \sum_{h=1}^{H} W_h \left(\bar{y}_h - \bar{y}\right)^2 = \left[\frac{1,182}{5,318} \times 11,783.6 + \ldots\right] + \left[\frac{1,182}{5,318} \times \left(72.7 - 2,583.4\right)^2 + \ldots\right] = 49,038,472.8$$

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A. Proportionate

- Considering

$$\sum_{h=1}^{H} n_h = n \qquad f_h = n_h / N_h \qquad f = n/N$$

- If $f_h = f$ for all strata, *epsem* sampling results
- Hence,

$$\frac{n_h}{N_h} = \frac{n}{N} \Rightarrow n_h = n \times \frac{N_h}{N} = n \times W_h$$

- Self-weighting design: No weighting needed for the mean
- Sampling variance is simplified:

$$\operatorname{var}\left(\bar{y}_w\right) = \left(\frac{1-f}{n}\right) \sum_{h=1}^{H} W_h s_h^2 = \left(\frac{1-f}{n}\right) s_w^2$$

Design effect

- Deff can be expresses as

$$deff = \frac{\operatorname{var}\left(\bar{y}_w\right)}{\operatorname{var}_{SRS}\left(\bar{y}\right)} = \frac{\left(\frac{1-f}{n}\right)s_w^2}{\left(\frac{1-f}{n}\right)s^2} = \frac{s_w^2}{s^2}$$

- An alternate expression shows how gains can be obtained:

$$deff = 1 - \frac{\sum_{h=1}^{H} W_h\left(\bar{y}_h - \bar{y}_w\right)^2}{s^2}$$

Example: Stratified sample data, *n* = 500

Stratum h	Group	N_h	W_h	n_h
1	< 50K sq. ft.	1,182	0.2223	111
2	50 - 250K	1,846	0.3471	174
3	250 - 4,000K	1,395	0.2623	131
4	>= 4,000K	895	0.1683	84
Total		5,318		500

$$W_1 = \frac{N_1}{N} = \frac{1{,}182}{5{,}318} = 0.2223$$

$$n_1 = n \times \frac{N_1}{N} = 500 \times 0.2223 = 111.132 \approx 111$$

Example: Stratified sample data, *n* = 500

Stratum h	Group	N_h	W_h	n_h	s_h^2
1	< 50K sq. ft.	1,182	0.2223	111	11,783.6
2	50 - 250K	1,846	0.3471	174	205,657.8
3	250 - 4,000K	1,395	0.2623	131	3,568,970.7
4	>= 4,000K	895	0.1683	84	183,039,437.2
Total		5,318		500	

$$\mathrm{var}(\bar{y}_w) = \left(\frac{1-f}{n}\right) s_w^2 = \left(\frac{1-\dfrac{500}{5,318}}{500}\right) \times [0.2223 \times 11,783.6 + \ldots] = 57,758.2$$

$$deff = \frac{\mathrm{var}(\bar{y}_w)}{\mathrm{var}_{SRS}(\bar{y})} = \frac{57,758.2}{\left(1-\dfrac{500}{5,318}\right)\dfrac{49,038,472.8}{500}} = 0.65$$

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B. Equal

- Consider $n_h = \dfrac{n}{H}$ and $f_h = (n/H)/N_h$
- Not *epsem* unless strata are the same size
- Estimation requires the full weighted expressions
- *deff* may actually be greater than 1
- Why use it?
  - Suppose $S_h = S_{h'}$ for two different strata $h \neq h'$
  - Equal allocation minimizes the sampling variance $\mathrm{var}\left(\bar{y}_h - \bar{y}_{h'}\right)$

Example: Stratified sample data, *n* = 500

Stratum h	Group	N_h	W_h	n_h	s_h^2
1	< 50K sq. ft.	1,182	0.2223	125	11,783.6
2	50 - 250K	1,846	0.3471	125	205,657.8
3	250 - 4,000K	1,395	0.2623	125	3,568,970.7
4	>= 4,000K	895	0.1683	125	183,039,437.2
Total		5,318		500	

$$\operatorname{var}(\bar{y}_w) = \sum_{h=1}^{H} W_h^2 \operatorname{var}(\bar{y}_h) = \sum_{h=1}^{H} W_h^2 \frac{1-f_h}{n_h} s_h^2$$

$$= \left[0.2223^2 \times \frac{1-\frac{125}{1,182}}{125} \times 11,783.6 \right] + \ldots + \left[0.1683^2 \times \frac{1-\frac{125}{895}}{125} \times 183,039,437.2 \right] = 49,038,472.8$$

Example: Stratified sample data, *n* = 500

Stratum h	Group	N_h	W_h	n_h	s_h^2
1	< 50K sq. ft.	1,182	0.2223	125	11,783.6
2	50 - 250K	1,846	0.3471	125	205,657.8
3	250 - 4,000K	1,395	0.2623	125	3,568,970.7
4	>= 4,000K	895	0.1683	125	183,039,437.2
Total		5,318		500	

$$deff = \frac{\mathrm{var}\left(\bar{y}_w\right)}{\mathrm{var}_{SRS}\left(\bar{y}\right)} = \frac{49{,}038{,}472.8}{\left(1-\dfrac{500}{5{,}318}\right)\dfrac{49{,}038{,}472.8}{500}} = 0.4238$$

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C. Neyman

- Consider $n_h = n \times W_h S_h \Big/ \sum_{h=1}^{H} W_h S_h$
- Smallest variance for $\bar{y}_w$
- Gains in precision greater than proportionate
- Need estimates of S_h
  - In practice, use reasonable estimates
  - Large gains require variation among S_h
  - Large gains unlikely for proportions
  - Values specific to Y
  - For multipurpose surveys, allocations will vary as S_h vary across characteristics

Example: Stratified sample data, *n* = 500

Stratum h	Group	N_h	W_h	s_h^2	n_h
1	< 50K sq. ft.	1,182	0.2223	11,783.6	4
2	50 - 250K	1,846	0.3471	205,657.8	27
3	250 - 4,000K	1,395	0.2623	3,568,970.7	84
4	>= 4,000K	895	0.1683	183,039,437.2	385
Total		5,318			500

$$n_1 = n \times \frac{W_1 s_1}{\sum_{h=1}^{4} W_h s_h} = 500 \times \frac{0.2223 \times \sqrt{11,783.6}}{0.2223 \times \sqrt{11,783.6} + \ldots + 0.1683 \times \sqrt{183,039,437.2}} = 4.084 \approx 4$$

Example: Stratified sample data, *n* = 500

Stratum h	Group	N_h	W_h	s_h^2	n_h
1	< 50K sq. ft.	1,182	0.2223	11,783.6	4
2	50 - 250K	1,846	0.3471	205,657.8	27
3	250 - 4,000K	1,395	0.2623	3,568,970.7	84
4	>= 4,000K	895	0.1683	183,039,437.2	385
Total		5,318			500

$$\mathrm{var}\left(\bar{y}_w\right)=\sum_{h=1}^{H} W_h^2 \mathrm{var}\left(\bar{y}_h\right)=\sum_{h=1}^{H} W_h^2 \frac{1-f_h}{n_h} s_h^2$$

$$=\left[0.2223^2 \times \frac{1-\frac{4}{1,182}}{4} \times 11,783.6\right]+\ldots+\left[0.1683^2 \times \frac{1-\frac{385}{895}}{385} \times 183,039,437.2\right]=11,470.2$$

Example: Stratified sample data, *n* = 500

Stratum h	Group	N_h	W_h	s_h^2	n_h
1	< 50K sq. ft.	1,182	0.2223	11,783.6	4
2	50 - 250K	1,846	0.3471	205,657.8	27
3	250 - 4,000K	1,395	0.2623	3,568,970.7	84
4	>= 4,000K	895	0.1683	183,039,437.2	385
Total		5,318			500

$$deff = \frac{\operatorname{var}(\bar{y}_w)}{\operatorname{var}_{SRS}(\bar{y})} = \frac{11,470.2}{\left(1 - \frac{500}{5,318}\right)\frac{49,038,472.8}{500}} = 0.1291$$

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D. Optimum

- Consider total fixed cost $C = \sum_{h=1}^{H} n_h J_h$
- Objective:
  - Minimize sampling variance for $\bar{y}_w$ subjected to total fixed cost *C*, **OR**
  - Minimize cost subjected to a specified sampling variance for $\bar{y}_w$
- Allocate $n_h = k W_h S_h \big/ \sqrt{J_h}$
  - *k* is a constant of proportionality
- Neyman allocation is a special case where costs are the same across strata

Optimum allocation - Steps

- For a total fixed cost *C:*
  - Compute preliminary allocation with $n_h^* = W_h S_h \big/ \sqrt{J_h}$
  - Calculate preliminary cost under that allocation: $C^* = \sum_{h=1}^{H} n_h^* J_h$
  - Compute constant of proportionality as $k = C \big/ C^*$
  - Adjust preliminary allocation $n_h = k \times n_h^*$
- Similar process for fixed sampling variance

Example: C = $150,000 for Optimum allocation

Stratum h	Group	W_h	s_h^2	J_h	n_h^*	n_h
1	< 50K sq. ft.	0.2223	11,783.6	100	2.413	7
2	50 - 250K	0.3471	205,657.8	100	15.742	43
3	250 - 4,000K	0.2623	3,568,970.7	200	35.041	97
4	>= 4,000K	0.1683	183,039,437.2	400	113.846	314
Total						461

$$n_1^* = W_1 s_1 \big/ \sqrt{J_1} = 0.2223 \times \sqrt{11{,}783.6} \big/ \sqrt{100} = 2.413$$

$$C^* = \sum_{h=1}^{H} n_h^* J_h = 2.413 \times 100 + 15.742 \times 100 + 35.041 \times 200 + 113.846 \times 400 = 54{,}362.08$$

$$k = C \big/ C^* = 150{,}000 / 54{,}362.08 = 2.759$$

$$n_1 = k \times n_1^* = 2.759 \times 2.413 = 6.657 \approx 7$$

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$$\operatorname{var}(\bar{y}_w) = \sum_{h=1}^{H} W_h^2 \operatorname{var}(\bar{y}_h) = \sum_{h=1}^{H} W_h^2 \frac{1-f_h}{n_h} s_h^2$$

$$= \left[0.2223^2 \times \frac{1-\dfrac{7}{1{,}182}}{7} \times 11{,}783.6 \right] + \ldots + \left[0.1683^2 \times \frac{1-\dfrac{314}{895}}{314} \times 183{,}039{,}437.2 \right] = 13{,}719.3$$

Example: *C* = \$150,000 for Optimum allocation

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$$deff = \frac{\text{var}\left(\bar{y}_w\right)}{\text{var}_{SRS}\left(\bar{y}\right)} = \frac{13,719.3}{\left(1 - \frac{461}{5,318}\right)\frac{49,038,472.8}{461}} = 0.1412$$

Unit 3: Stratification
Stratification Topics

A. Number of strata
B. Paired selection
C. Collapsed strata
D. Poststratification
E. Sub-classes
F. Step-by-step selection

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A. Number of strata

- Stratification requires discrete categories
  - Stratifying variables may be discrete
  - Continuous stratifying variables divided into categories

- How many categories to capture gains possible?
  - Generally 3-6 strata adequate
  - When more than one stratifier, "coarser" cuts on more variables preferred to "finer" cuts
  - "Deepest" stratification for $n_h = 2$ (or $H = n/2$)
  - "Even deeper" stratification: 1 per stratum

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B. Paired selection

- Paired selections n_h = 2 useful in practice
  - "Deepest" stratification possible that allows sampling variances to be estimated without assumptions
  - Paired selection is *epsem* when: $N_h = N/H$
  - Attraction of paired selection is the simplification in variance estimation
  - When the design is *epsem*, proportionately allocated

Paired selection estimation

$$\bar{y} = \frac{\sum_{h=1}^{H}\sum_{i=1}^{n_h} y_{hi}}{n} = \frac{\sum_{h=1}^{H}\left(y_{h1} + y_{h2}\right)}{n}$$

$$\operatorname{var}\left(\bar{y}\right) = \left(\frac{1-f}{n}\right)\sum_{h=1}^{H} W_h s_h^2 \quad \left(where\ W_h = \frac{N_h}{N} = \frac{n_h}{n} = \frac{2}{n}\right)$$

$$= \left(\frac{1-f}{n}\right)\sum_{h=1}^{H}\frac{2}{n}\left[\left(y_{h1} - \frac{y_{h1}+y_{h2}}{2}\right)^2 + \left(y_{h2} - \frac{y_{h1}+y_{h2}}{2}\right)^2\right]$$

$$= 2\left(\frac{1-f}{n^2}\right)\sum_{h=1}^{H}\left[\left(\frac{y_{h1}-y_{h2}}{2}\right)^2 + \left(\frac{y_{h2}-y_{h1}}{2}\right)^2\right]$$

$$= \left(\frac{1-f}{n^2}\right)\sum_{h=1}^{H}\left(y_{h1} - y_{h2}\right)^2$$

Paired selection features

- For element sampling, the symmetry of the selection and variance estimation disrupted by
  - Blanks in the list, non-responding elements
  - Analysis of subclasses
- Remedy: collapse strata
  - Increases the number of selections within "collapsed" or "pseudo" strata
  - Sampling variances are estimated as though the stratum boundary(ies) do not exist
  - “Deeper” stratification, which can increase the precision of estimates, is ignored
  - Sampling variance is over-estimated.

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C. Collapsed strata

Stratum	Elements	Collapsed stratum	Elements
1	y_{11}, y_{12}	1	$y_{11}, y_{12}, y_{21}, y_{22}$
2	y_{21}, y_{22}		
3	y_{31}, y_{32}	2	$y_{31}, y_{32}, y_{41}, y_{42}$
4	y_{41}, y_{42}		

Collapsed strata properties

- Used to simplify more complicated designs
- Reduced number of strata in estimation
- Increased degrees of freedom (n – H)
- Variance estimate based on collapsed strata biased for variance for original strata
  - Over-estimates variance by factor proportionate to between stratum difference

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D. Poststratification

- Poststratification: variables to be used to create strata are not available at the time of selection
  - Stratify after selection using variables collected during the survey
  - Gains in precision are possible, with suitable modification to variance estimation
  - Population control adjustment
  - Poststratification requires
    - Poststrata known for each element
    - Poststratum weights W_g for each poststratum

Poststratified estimator

- Sort the sample into poststrata
- Compute poststratum means
- Compute poststratified mean as $\bar{y}_{ps} = \sum_{g=1}^{G} W_g \bar{y}_g$
- For the variance,

$$Var\left(\bar{y}_{ps}\right) \approx \left(\frac{1-f}{n}\right)\left[\sum_{g=1}^{G} W_g S_g^2 + \sum_{g=1}^{G} W_g \left(1-W_g\right)\frac{S_g^2}{n_g}\right]$$

Poststratification illustration

- Sample n=2,000 SRS from N=20,000 libraries
  - g=1: ordered last year and prior years
  - g=2: ordered last year only
  - g=3: never ordered

g	W_g	n_g	n_g/n	$\bar{y}_g$	s_g^2
1	0.20	410	0.205	1250	3.2×10^6
2	0.25	493	0.247	750	3.0×10^6
3	0.55	1097	0.548	140	1.5×10^6
		2000	1.000		($s^2 = 2.4 \times 10^6$)

Estimation

- The mean and variance are respectively

$$\bar{y}_{ps} = (0.2)(1250) + (0.25)(750) + (0.55)(140) = 514.5$$

$$\text{var}(\bar{y}_{ps}) = \left(\frac{1-0.1}{2000}\right)\left[(0.2)(3.2\times10^6) + (0.2)(1-0.2)\frac{3.2\times10^6}{410} + \cdots\right] = 998.0$$

- Under proportionate allocation

$$\text{var}(\bar{y}) = \left(\frac{1-f}{n}\right)\sum_{g=1}^{G} W_g s_g^2 = 996.8$$

- Under SRS

$$\text{var}_{SRS}(\bar{y}) = \left(\frac{1-f}{n}\right)s^2 = 1086.4$$

Poststratification gains in precision

- The design effects are …

$$deff_1 = \frac{\text{var}\left(\bar{y}_{ps}\right)}{\text{var}_{SRS}\left(\bar{y}\right)} = 0.9575 \qquad deff_2 = \frac{\text{var}\left(\bar{y}\right)}{\text{var}_{SRS}\left(\bar{y}\right)} = 0.9563$$

- W_g may be approximate
- Use of the W_g and poststratified estimators are a bias reduction technique
- Multipurpose poststratification limits choice of W_g
- Variance estimate: Taylor series approximation

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E. Sub-classes

- Stratification gains diminish for subclasses
- Let M = total number of subclass elements in the population
- M_h denotes the number for stratum h
- m = subclass sample size (not "controlled")
- m_h is the size for stratum h
- The relative size of the subclass is given by

$$\bar{M} = \frac{M}{N} \text{ and } \bar{m} = \frac{m}{n}$$

Stratified sub-class estimator

- Consider the subclass mean for stratified random sampling
- The individual M_h will not be known
- Estimate M_h from the sample as : $\hat{M}_h = F_h m_h = \frac{N_h}{n_h} m_h$

Losses in gains in precision

- Subclass mean "loses" gains in precision
- For large subclasses, 1 - $\bar{m}_h$ small, gains retained
- For small subclasses, 1 - $\bar{m}_h$ = 1, gains lost
- Resulting estimate no more precise than an SRS of the subclass
- Loss is approximately

$$\left(1-m_h^{'}\right)\sum_{h=1}^{H}\left(W_h^{'}\right)^2\left(\bar{y}_h-\bar{y}_w\right)^2$$

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F. Step-by-step

- The "steps" in stratified sampling are as follows:
  - Identify stratifying variables correlated with the measure(s) of interest
  - Choose “cuts" on the stratifying variables
  - Divide the population into strata
  - S_h, *deff*, and S^2 are known, at least approximately, and that *fpc*’s within strata can be ignored
  - Compute an SRS sample size $n = n_{SRS} \times deff$

Finishing the design

- Determine an allocation for the desired n
  - Proportionate
  - Neyman or optimum
  - Equal
- Adjust SRS n based on expected *deff* & allocate
- Select sample and compute estimates taking the stratified sample selection into account
  - Mean
  - Variance